

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital VLSI Testing (01CT0629)	Aim: To model and detect stuck-at-0 and stuck-at-1 faults in D flipflop and SR Flipflop.	
Experiment No: 06	Date:13/03/2026	Enrolment No: 92301733056

AIM: To model and detect stuck-at-0 and stuck-at-1 faults in D flipflop and SR Flipflop.

IDE: EDA Playground

THEORY:

Stuck-At Fault Model

In digital circuits, stuck-at faults occur when a signal line is permanently fixed to logic 0 or 1, regardless of the intended circuit operation.

There are two types:

Stuck-at-0 (SA0)

The signal line is permanently fixed to logic 0.

Stuck-at-1 (SA1)

The signal line is permanently fixed to logic 1.

These faults are commonly used in VLSI testing to detect defects in circuits.

D Flip-Flop

A D Flip-Flop (Data Flip-Flop) stores one bit of data and transfers input D to output Q on the positive edge of the clock.

Characteristic Equation

$$Q_{next} = D$$

If faults occur:

Fault	Effect
SA0	Output always 0
SA1	Output always 1

SR Flip-Flop

The SR Flip-Flop (Set-Reset) has two inputs:

S → Set

R → Reset

Operation Table

S	R	Q(next)
0	0	No change
0	1	0
1	0	1
1	1	Invalid

Fault Behavior

Fault	Effect
SA0	Signal always 0
SA1	Signal always 1

DESIGN AND IMPLEMENTATION:

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VERILOG CODE:

A. D flipflop Module

```

module d_flipflop_fault(
    input clk,
    input reset,
    input d,
    input sa0,    // stuck-at-0 fault
    input sa1,    // stuck-at-1 fault
    output reg q
);

always @(posedge clk or posedge reset)
begin
    if(reset)
        q <= 0;

    else if(sa0)
        q <= 0;    // output stuck at 0

    else if(sa1)
        q <= 1;    // output stuck at 1

    else
        q <= d;    // normal operation
end

endmodule

```

B. Testbench

```

// Code your testbench here
// or browse Examples
`timescale 1ns/1ps

module tb_d_flipflop;

    reg clk;
    reg reset;
    reg d;
    reg sa0;
    reg sa1;

    wire q;

    d_flipflop_fault uut (
        .clk(clk),
        .reset(reset),
        .d(d),

```

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```

        .sa0(sa0),
        .sa1(sa1),
        .q(q)
    );

    always #5 clk = ~clk;

    initial
    begin
        $dumpfile("dump.vcd");
        $dumpvars(0,tb_d_flipflop);
        clk = 0;
        reset = 1;
        d = 0;
        sa0 = 0;
        sa1 = 0;

        #10 reset = 0;

        // Normal operation
        #10 d = 1;
        #10 d = 0;
        #10 d = 1;

        // Stuck-at-0 fault
        #10 sa0 = 1;
        #20 sa0 = 0;

        // Stuck-at-1 fault
        #10 sa1 = 1;
        #20 sa1 = 0;

        #40 $finish;
    end

endmodule

```

SIMULATION RESULTS:



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SR FLIP_FLOP :

DESIGN CODE :

```

module sr_flipflop_fault(
    input clk,
    input reset,
    input s,
    input r,
    input sa0,    // stuck-at-0 fault
    input sa1,    // stuck-at-1 fault
    output reg q
);

always @(posedge clk or posedge reset)
begin
    if(reset)
        q <= 0;

    else if(sa0)
        q <= 0;    // output stuck at 0

    else if(sa1)
        q <= 1;    // output stuck at 1

    else
    begin
        if(s==0 && r==0)
            q <= q; // no change
        else if(s==0 && r==1)
            q <= 0; // reset
        else if(s==1 && r==0)
            q <= 1; // set
    end
end

endmodule

```

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TESTBENCH CODE ;

```

`timescale 1ns/1ps

module tb_sr_flipflop;

reg clk;
reg reset;
reg s;
reg r;
reg sa0;
reg sa1;

wire q;

sr_flipflop_fault uut (
    .clk(clk),
    .reset(reset),
    .s(s),
    .r(r),
    .sa0(sa0),
    .sa1(sa1),
    .q(q)
);

always #5 clk = ~clk;

initial
begin
$dumpfile("dump.vcd");
$dumppvars(0,tb_sr_flipflop);
    clk = 0;
    reset = 1;
    s = 0;
    r = 0;
    sa0 = 0;
    sa1 = 0;

    #10 reset = 0;

    // Normal operation
    #10 s = 1; r = 0;
    #10 s = 0; r = 1;
    #10 s = 0; r = 0;

    // Stuck-at-0 fault
    #10 sa0 = 1;
    #20 sa0 = 0;

    // Stuck-at-1 fault
    #10 sa1 = 1;
    #20 sa1 = 0;

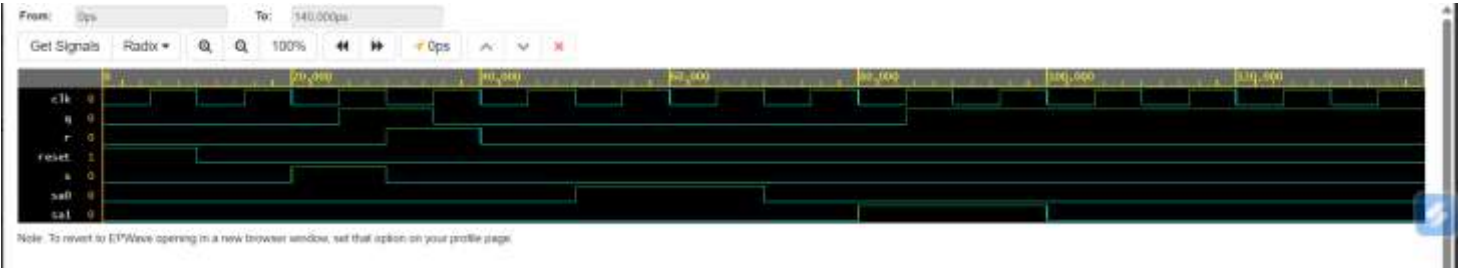
    #40 $finish;

```

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```
end
endmodule
```

SIMULATION RESULTS :



CONCLUSION:

In this experiment, D Flip-Flop and SR Flip-Flop circuits were designed and simulated using Verilog HDL. Stuck-at fault models, namely stuck-at-0 (SA0) and stuck-at-1 (SA1), were introduced to analyze the behavior of the circuits under faulty conditions. The simulation results showed that when a stuck-at fault is activated, the output becomes permanently fixed at logic 0 or logic 1, regardless of the normal input conditions.

This experiment demonstrates how fault modeling helps in identifying defects in digital circuits during testing and verification. Therefore, the implementation of stuck-at fault detection improves the reliability and testing efficiency of sequential circuits such as D Flip-Flops and SR Flip-Flops.